

Lab 2 Worksheet Report

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GitHub Repository: (to be added)

W&B Project Link:

<https://wandb.ai/b23bb1043-indian-institute-of-technology-jodhpur/CIFAR10-CNN-Lab2/runs/nhiea507?nw=nwuserb23bb1043>

Google colab -

https://colab.research.google.com/drive/19gvqqAYHWdlEsyRx2rPC_vjfqTF3P0RX#scrollTo=Kb8tNg4WWSnj

Aim -

to design and train a Convolutional Neural Network (CNN) on the CIFAR-10 dataset using PyTorch , also creating a custom data loader, training the model for multiple epochs, computing model complexity in terms of parameters and FLOPs, visualizing gradient flow and weight update flow, and reporting all results using Weights & Biases.

Dataset -

The CIFAR-10 dataset consists of 60,000 color images of size $32 \times 32 \times 3$, belonging to 10 classes.

The dataset is split into 50,000 training images and 10,000 test images.

Custom DataLoader -

A custom DataLoader class was implemented to load the CIFAR-10 dataset and apply appropriate transformations.

1. Training Transformations

- RandomCrop (with padding)
- RandomHorizontalFlip
- Normalization

These augmentations increase data diversity and help reduce overfitting.

2. Test Transformations

- Only normalization

This ensures fair evaluation without introducing randomness during testing.

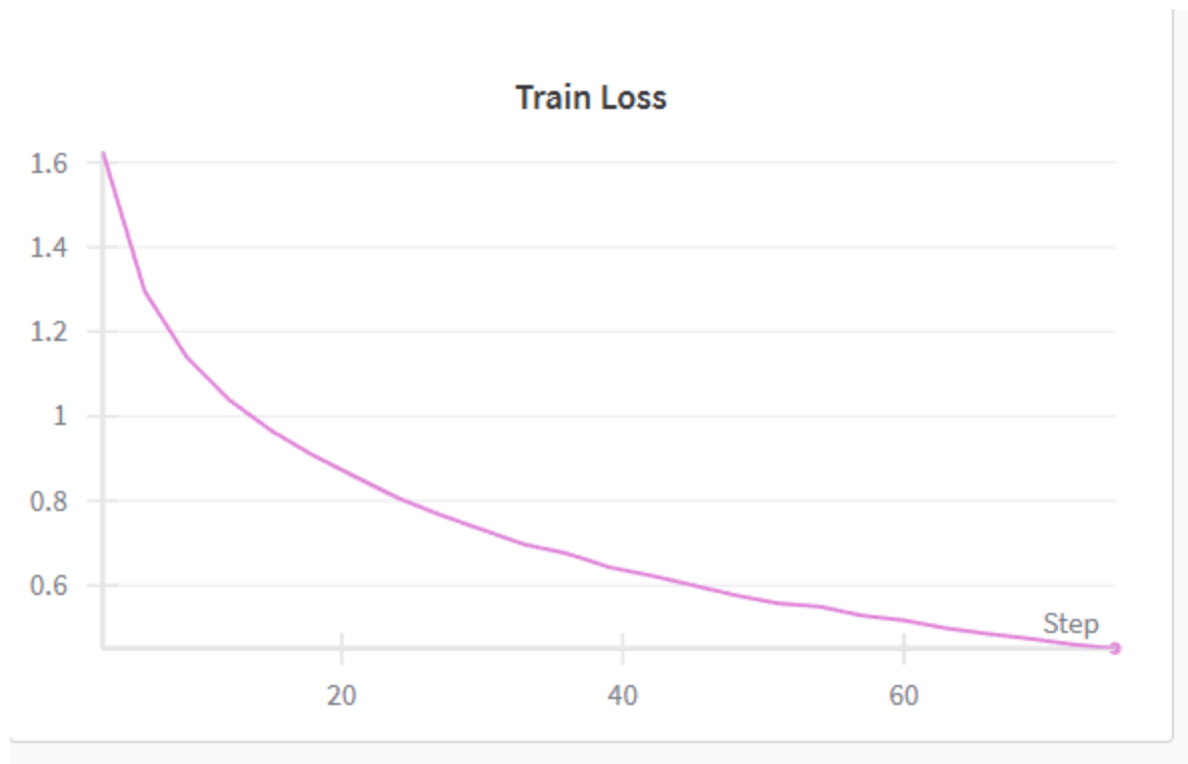
Model Architecture -

A custom lightweight VGG-style CNN was designed for this experiment.

Model Complexity:

- Total Parameters: ~2.47 Million
- FLOPs: ~42 MFLOPs

Training Setup -



- Loss Function: CrossEntropyLoss
- Optimizer: Adam
- Learning Rate: $1e-3$
- Batch Size: 64
- Epochs: 25
- Device: GPU

The model was trained for 25 epochs, and training loss was logged at the end of each epoch.

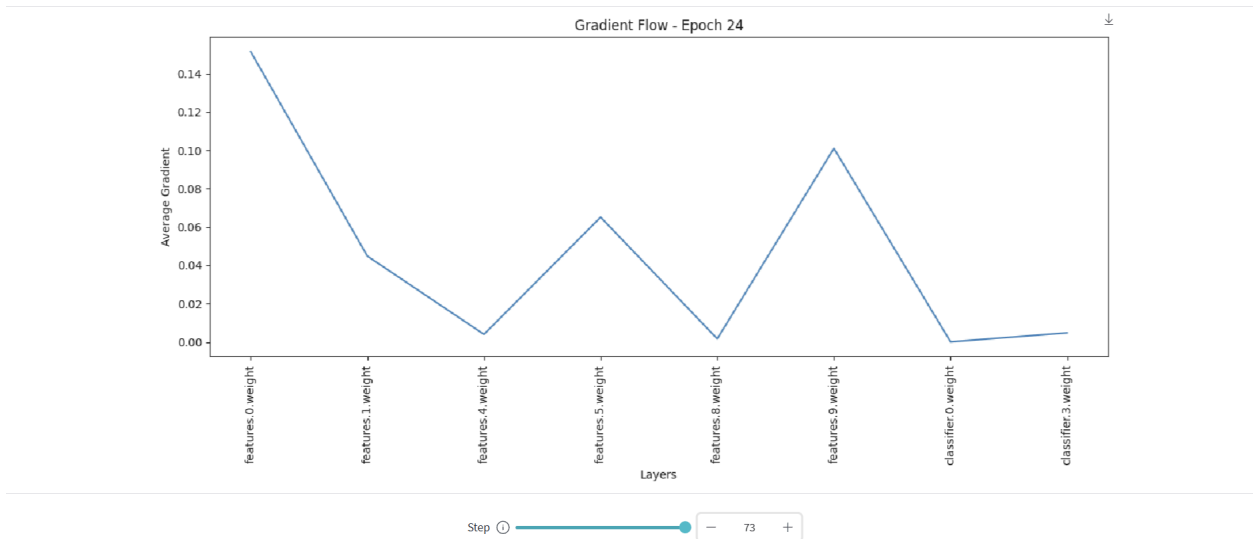
FLOPs and Parameter Counting -

Model complexity was computed using a FLOPs-counting utility. Since the architecture remains fixed during training:

- FLOPs and parameter count remain constant across epochs
- These values were logged once to Weights & Biases

Visualization and Analysis

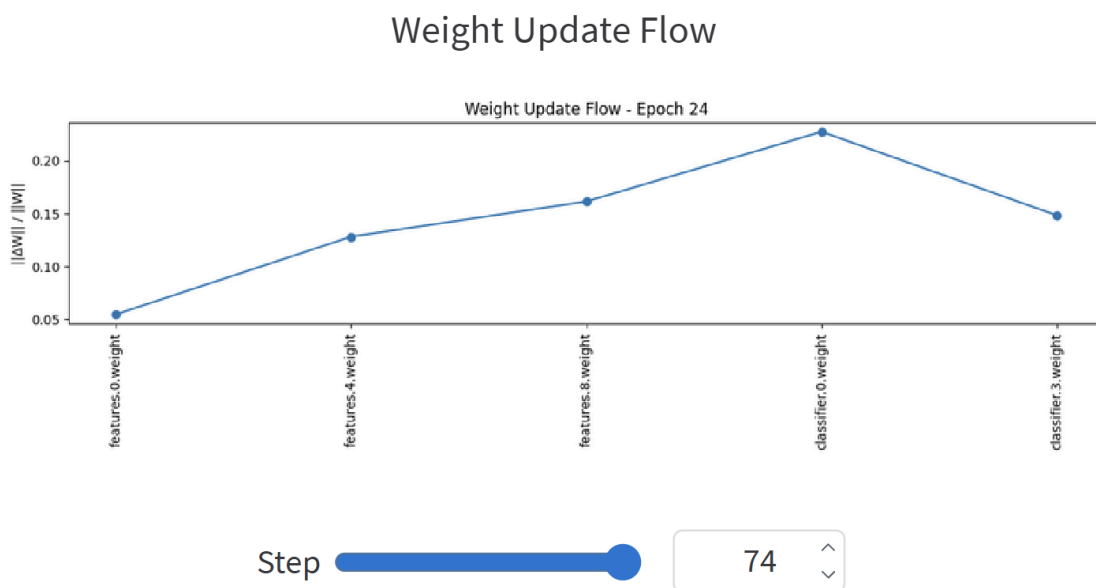
Gradient Flow Visualization



Observation:

- Gradients are non-zero across most layers, indicating effective backpropagation.
- Early convolution layers have smaller gradients compared to deeper layers, which is expected.
- No evidence of vanishing or exploding gradients is observed.
- Gradient magnitudes remain stable in later epochs, showing stable training.

Weight Update Flow Visualization



- Weight updates are larger in deeper layers (classifier and last convolution layers).
- Earlier layers show smaller update ratios, indicating gradual feature refinement.
- Update magnitudes decrease towards later epochs, showing convergence behavior.
- No abnormal spikes are observed, suggesting stable and well-controlled learning.

Weights & Biases Logging

All metrics and visualizations were logged using Weights & Biases, including:

- Training loss per epoch
- FLOPs and parameter count
- Gradient flow images
- Weight update flow images

Results -

Metric	Value
Epochs	25
Parameters	~2.47M
FLOPs	~42 MFLOPs
Final Training Loss	~0.43